Vector Spaces

DEFINITION: Let V be a nonempty set with two operations:

(i) ***Vector Addition:*** This assigns to any $u, v \in V$ a *sum* $u + v$ in V.

(ii) ***Scalar Multiplication:*** This assigns to any $u \in V$, $k \in K$ a *product* $ku \in V$.

Then V is called a *vector space* (over the field K) if the following axioms hold for any vectors $u, v, w \in V$:

[A_1] $(u + v) + w = u + (v + w)$

[A_2] There is a vector in V, denoted by 0 and called the *zero vector*, such that, for any $u \in V$,

$$u + 0 = 0 + u = u$$

[A_3] For each $u \in V$, there is a vector in V, denoted by $-u$, and called the *negative* of u, such that

$$u + (-u) = (-u) + u = 0.$$

[A_4] $u + v = v + u$.

[M_1] $k(u + v) = ku + kv$, for any scalar $k \in K$.

[M_2] $(a + b)u = au + bu$, for any scalars $a, b \in K$.

[M_3] $(ab)u = a(bu)$, for any scalars $a, b \in K$.

[M_4] $1u = u$, for the unit scalar $1 \in K$.

Examples of vector space:

1. $\mathbb{R}^2 = \{(a, b): a, b \in \mathbb{R}\}$ is a vector space over the field $K = \mathbb{R}$ with respect to the operations of vector addition and scalar multiplication on $\mathbb{R}^2$.

2. $\mathbb{R}^3 = \{(a, b, c): a, b \in \mathbb{R}\}$ is a vector space over the field $K = \mathbb{R}$ with respect to the operations of vector addition and scalar multiplication on $\mathbb{R}^3$.

3. *The set V of all 2×2 matrices with real entries is a vector space if addition is defined to be matrix addition and scalar multiplication is defined to be matrix multiplication. (Solution to be done in class).*

Example: Let $V = \mathbb{R}^2$ and define addition and scalar multiplication operations as follows: If $\boldsymbol{u} = (u_1, u_2)$ and $\boldsymbol{v} = (v_1, v_2)$, then define $\boldsymbol{u} + \boldsymbol{v} = (u_1 + v_1, u_2 + v_2)$ and if k is any real number, then define $k\boldsymbol{u} = (ku_1, 0)$. Show that V is not a vector space.

Solution: If $\boldsymbol{u} = (u_1, u_2)$ is such that $u_2 \neq 0$, then $1\,\boldsymbol{u} = 1(u_1, u_2) = (1.u_1, 0) = (u_1, 0) \neq \boldsymbol{u}$ **.** Therefore the last axiom is not satisfied. Thus V is not a vector space with the stated operations.

Exercise: Show that the set of all 2×2 matrices of the form $\begin{bmatrix} a & 1 \\ 1 & b \end{bmatrix}$ with addition defined by $\begin{bmatrix} a & 1 \\ 1 & b \end{bmatrix} + \begin{bmatrix} c & 1 \\ 1 & d \end{bmatrix} = \begin{bmatrix} a + c & 1 \\ 1 & b + d \end{bmatrix}$ and scalar multiplication defined by $k\begin{bmatrix} a & 1 \\ 1 & b \end{bmatrix} = \begin{bmatrix} ka & 1 \\ 1 & kb \end{bmatrix}$ is a vector space. What is the zero vector in this space?

Subspace: A subset W of a vector space V is called a **subspace** of V if W is itself a vector space under the addition and scalar multiplication defined on V.

First Fundamental Theorem of Subspace: If W is a set of one or more vectors from a vector space V, then W is a subspace of V if and only if the following conditions hold.

a) If $\boldsymbol{u}$ and $\boldsymbol{v}$ are vectors in W, then $\boldsymbol{u}+\boldsymbol{v}$ is in W.
b) If k is any scalar and $\boldsymbol{u}$ is any vector in W, then $k\boldsymbol{u}$ is in W.

Second Fundamental Theorem of Subspace: W is a subspace of V if and only if the following conditions hold.

a) $0 \in W$ or $W \neq \emptyset$
b) $\boldsymbol{u}, \boldsymbol{v} \in W \Rightarrow \alpha\boldsymbol{u}+\beta\boldsymbol{v} \in W$ where α, β are scalars.

Example: Show that any straight line passing through origin is a subspace of $\mathbb{R}^2$.

Solution: Let the set of any straight line passing through origin is

$$S=\{(x,y): y=mx\}$$

Now we have to show that S is a subspace of $\mathbb{R}^2$.

$\mathbf{0}=(0,0) \in S$ since $(0,0)$ satisfies $y=mx$, so that S is non-empty.

Let $\boldsymbol{u}, \boldsymbol{v} \in S$ where $\boldsymbol{u}=(u_1,u_2)$ and $\boldsymbol{v}=(v_1,v_2)$

$\therefore u_2=mu_1$ and $v_2=mv_1$

For any scalar α and β, we have

$$\begin{aligned}\alpha\boldsymbol{u}+\beta\boldsymbol{v} &= \alpha(u_1,u_2)+\beta(v_1,v_2)\\ &=(\alpha u_1,\alpha u_2)+(\beta v_1,\beta v_2)\\ &=(\alpha u_1+\beta v_1, \alpha u_2+\beta v_2)\\ &=(w_1,w_2)\end{aligned}$$

where $w_1=\alpha u_1+\beta v_1, w_2=\alpha u_2+\beta v_2$

Here, $w_2=\alpha u_2+\beta v_2=\alpha.mu_1+\beta.mv_1=m(\alpha u_1+\beta v_1)=mw_1$

$$\therefore (w_1,w_2) \in S \Rightarrow \alpha\boldsymbol{u}+\beta\boldsymbol{v} \in S$$

Thus S is a subspace of $\mathbb{R}^2$.

Hence any straight line passing through origin is a subspace of $\mathbb{R}^2$.

Example: Geometrically show that the points in a plane through the origin of $\mathbb{R}^3$ is a subspace of $\mathbb{R}^3$.

Solution: Let W be any plane through the origin, and let $\boldsymbol{u}$ and $\boldsymbol{v}$ be any vectors in W. Then $\boldsymbol{u}+\boldsymbol{v}$ must lie in W because it is the diagonal of the parallelogram determined by $\boldsymbol{u}$ and $\boldsymbol{v}$ (Figure 1), and $k\boldsymbol{u}$ must lie in W for any scalar k because $k\boldsymbol{u}$ lies on a line through $\boldsymbol{u}$. Thus W is closed under addition and scalar multiplication, so it is a subspace of $\mathbb{R}^3$.

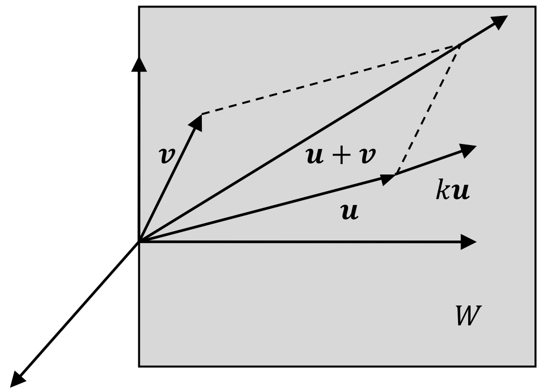

Figure: 1

Example: Let W be the set of all points (x, y) in $\mathbb{R}^2$ such that $x \geq 0, y \geq 0.$ Show that the set W is not a subspace of $\mathbb{R}^2$.

Solution: W is not closed under scalar multiplication. For example, $(1,1)$ lies in W, but its negative $(-1)\boldsymbol{v} = -\boldsymbol{v} = (-1,-1) \notin W$. Therefore, the set W is not a subspace of $\mathbb{R}^2$.

Exercise:

1. Show that $T = \{(a, b, c, d) \in \mathbb{R}^4 : 2a - 3b + 5c - d = 0\}$ is a subspace of $\mathbb{R}^4$.
2. Show that $W = \{(a, b, c) : a, b, c \in \mathbb{R}$ and $a - 2b + 3c = 5\}$ is not a subspace of $\mathbb{R}^3$.

Linear Combination:

A vector $\boldsymbol{w}$ is called a **linear combination** of the vectors $\boldsymbol{v_1}, \boldsymbol{v_2}, \dots \dots \dots, \boldsymbol{v_r}$ if it can be expressed in the form

$$\boldsymbol{w} = k_1\boldsymbol{v_1} + k_2\boldsymbol{v_2} + \cdots\cdots\cdots + k_r\boldsymbol{v_r}$$

where $k_1, k_2, \dots \dots \dots, k_r$ are scalars.

Example: Consider the vectors $\boldsymbol{u} = (1,2,-1)$ and $\boldsymbol{v} = (6,4,2)$ in $\mathbb{R}^3$. Show that $\boldsymbol{w} = (9,2,7)$ is A linear combination of $\boldsymbol{u}$ and $\boldsymbol{v}$ and that $\boldsymbol{w'} = (4,-1,8)$ is not a linear combination of $\boldsymbol{u}$ and $\boldsymbol{v}$.

Solution: In order for $\boldsymbol{w}$ to be a linear combination of $\boldsymbol{u}$ and $\boldsymbol{v}$**,** there must be scalars k_1 and k_2 such that $\boldsymbol{w} = k_1\boldsymbol{u} + k_2\boldsymbol{v}$ **;** that is**,**

$$(9,2,7) = k_1(1,2,-1) + k_2(6,4,2)$$
$$\Rightarrow (9,2,7) = (k_1 + 6k_2, 2k_1 + 4k_2, -k_1 + 2k_2)$$

Equating corresponding components gives

$$k_1 + 6k_2 = 9$$
$$2k_1 + 4k_2 = 2$$
$$-k_1 + 2k_2 = 7$$

Solving this system using Gaussian elimination

$$\sim \begin{cases} k_1 + 6k_2 = 9 \\ \quad 8k_2 = 16 \\ \quad 8k_2 = 16 \end{cases}$$
$$[L_2 \to 2L_1 - L_2, L_3 \to L_1 + L_3]$$

$$\sim \begin{cases} k_1 + 6k_2 = 9 \\ \quad k_2 = 2 \end{cases}$$

Therefore $k_1 = -3$, $k_2 = 2$ so that

$$\boldsymbol{w} = -3\boldsymbol{u} + 2\boldsymbol{v}$$

Similarly, for $\boldsymbol{w}'$ to be linear combination of $\boldsymbol{u}$ and $\boldsymbol{v}$**,** there must be scalars k_1 and k_2 such that $\boldsymbol{w}' = k_1\boldsymbol{u} + k_2\boldsymbol{v}$ **;** that is,

$$(4,-1,8) = k_1(1,2,-1) + k_2(6,4,2)$$
$$\Rightarrow (4,-1,8) = (k_1 + 6k_2, 2k_1 + 4k_2, -k_1 + 2k_2)$$

Equating corresponding components gives

$$k_1 + 6k_2 = 4$$
$$2k_1 + 4k_2 = -1$$
$$-k_1 + 2k_2 = 8$$

Solving this system using Gaussian elimination

$$\sim \begin{cases} k_1 + 6k_2 = 4 \\ \quad 8k_2 = 9 \\ \quad 8k_2 = 12 \end{cases}$$
$$[L_2 \to 2L_1 - L_2, L_3 \to L_1 + L_3]$$

$$\sim \begin{cases} k_1 + 6k_2 = 4 \\ \quad 8k_2 = 9 \\ \quad 0 = -3 \end{cases}$$
$$[L_3 \to L_2 - L_3]$$

This implies that the given system is inconsistent, so no such scalars k_1 and k_2 exist. Consequently, $\boldsymbol{w}'$ is not a linear combination of $\boldsymbol{u}$ and $\boldsymbol{v}$**.**

Example: Write the matrix $A = \begin{bmatrix} 3 & -1 \\ 1 & -2 \end{bmatrix}$ as a linear combination of the matrices $A_1 = \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix}$, $A_2 = \begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix}$ and $A_3 = \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}$.

Solution: Set A as a linear combination of A_1, A_2 and A_3 using the unknowns k_1, k_2 and k_3.

$$A = k_1A_1 + k_2A_2 + k_3A_3$$

That is,

$$\begin{bmatrix} 3 & -1 \\ 1 & -2 \end{bmatrix} = k_1\begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix} + k_2\begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix} + k_3\begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 3 & -1 \\ 1 & -2 \end{bmatrix} = \begin{bmatrix} k_1 & k_1 \\ 0 & -k_1 \end{bmatrix} + \begin{bmatrix} k_2 & k_2 \\ -k_2 & 0 \end{bmatrix} + \begin{bmatrix} k_3 & -k_3 \\ 0 & 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 3 & -1 \\ 1 & -2 \end{bmatrix} = \begin{bmatrix} k_1 + k_2 + k_3 & k_1 + k_2 - k_3 \\ -k_2 & -k_1 \end{bmatrix}$$

Equating corresponding components and forming the linear system we get

$$\begin{aligned} k_1 + k_2 + k_3 &= 3 \\ k_1 + k_2 - k_3 &= -1 \\ -k_2 \qquad &= 1 \\ -k_1 \qquad\qquad &= -2 \end{aligned}$$

Hence the solution of the system is $k_1 = 2, k_2 = -1, k_3 = 2$

Therefore,

$$A = 2A_1 - A_2 + 2A_3$$

That is A as a linear combination of A_1, A_2 and A_3.

Exercise Problems:

1. Consider the vectors $\boldsymbol{v_1} = (2,1,4), \boldsymbol{v_2} = (1,-1,3)$ and $\boldsymbol{v_3} = (3,2,5)$ in $\mathbb{R}^3$. Show that $\boldsymbol{v} = (5,9,5)$ is a linear combination of $\boldsymbol{v_1}, \boldsymbol{v_2}$ and $\boldsymbol{v_3}$**.**

   Answer: $\boldsymbol{v} = 3\boldsymbol{v_1} - 4\boldsymbol{v_2} + \boldsymbol{v_3}$.

2. For which value of λ the vector $\boldsymbol{v} = (1, \lambda, 5)$ in $\mathbb{R}^3$ will be a linear combination of the vectors $\boldsymbol{v_1} = (1,-3,2)$ and $\boldsymbol{v_2} = (2,-1,1)$.

   Answer: $\lambda = -8$ and $k_1 = 3, k_2 = -1$.